SIKSHA ‘O’ ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 44 **GROUP NO: 14**

PROJECT TITLE: A Modular Framework for Prompt Injection Detection in GPT-Based AI Systems

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Problem to be Addressed:

GPT-based AI systems are now widely used in organizations as well as public platforms. Even though these systems include built-in safety measures, they can still be vulnerable to prompt injection attempts. In such cases, users try to manipulate the AI’s instructions or extract unintended information. The risk becomes more important in enterprise environments where AI systems are connected to internal documents and workflows. In these situations, a manipulated prompt may increase the chances of unintended data exposure. Therefore, there is a need for additional input-level analysis to improve reliability and control.

Aim and Objectives:

The aim of this project is to develop a modular prompt analysis framework that can identify potentially harmful or manipulated prompts before they are processed by a GPT-based system. The objectives include creating a labeled dataset of normal and suspicious prompts, training machine learning and transformer-based models, evaluating their performance, and developing a risk scoring approach to assess prompt safety.

Functionalities and Technical Approach:

The system will analyze incoming prompts and estimate their likelihood of being injection attempts using trained models. A risk probability will be generated for each prompt. Based on experimental validation, either a probability threshold based decision or a classification-based approach will be used for filtering. The final approach will be selected after comparative evaluation using metrics such as accuracy, precision, recall, and F1-score. The chosen model will be integrated as an API-based screening module that works before the GPT system processes the input.

Benefits and Social Contribution:

This project supports safer AI deployment by adding an additional prompt-level analysis layer. It focuses on improving reliability without making unrealistic claims about completely eliminating security risks.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

HARDWARE SPECIFICATIONS:

- Processor: Intel i5 / Ryzen 7
- RAM: 16 GB
- Storage: Sufficient available storage space
- GPU: NVIDIA GeForce RTX 3050

METHODS / ALGORITHMS:

- The project will use supervised learning techniques for prompt classification. TF-IDF based feature extraction with Logistic Regression and Support Vector Machine will be implemented as baseline models. A lightweight transformer model such as DistilBERT will be fine-tuned for contextual prompt analysis. The trained model will generate a risk probability score for each prompt. Based on validation results, a filtering mechanism will be implemented to control whether a prompt should be forwarded to the GPT-based system.
- Performance evaluation will be conducted using standard metrics including accuracy, precision, recall, and F1-score to ensure balanced detection performance.

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

Name	Registration Number	Signature
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(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

Mohinikanta Sahoo

Project Supervisor	Section Coordinator, FRP	FRP Coordinator